

Supporting Information

Distance dependent interaction between a single emitter and a single dielectric nanoparticle using DNA origami

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Calculations of silicon NP size distribution. We estimate the dispersity of our SiNP solution using the measured extinction spectrum. We then tried to reproduce the measured extinction spectrum from the calculated spectra of single SiNPs by assuming a normal size distribution with average diameter $\bar{D}$ and standard deviation σ as fitting parameters.^[48] The measured and calculated spectra are shown in **Figure 1C**. The best fit obtained shows good agreement with the values obtained from TEM measurements, indicating high shape uniformity and dispersion. Under the Beer-Lambert law (i.e., no multiple scattering), extinction can be described as

$$Extinction = \varepsilon lc \sim 0.434 \sigma_{ext} lN$$

where ε is the extinction coefficient, l the path length, c the concentration, and N the number of NPs. In a solution with a size distribution, the extinction cross-section σ_{ext} can be written as a function of the NP's diameter D and the probability density function P as

$$\sigma_{ext} = \int_0^{\infty} \sigma_e(D, \lambda) P(D) dD$$

where σ_e is the size and wavelength-dependent extinction cross section, obtained from simulations and used to compute the simulated extinction spectrum.

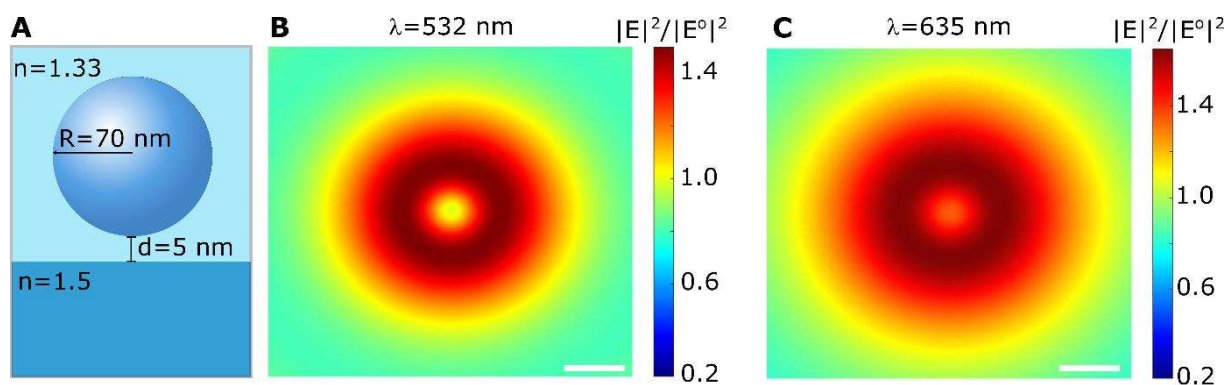


Figure S1: Electric field enhancement at the emitter position plane. Maps of the enhancement of the intensity of the electric field at the transversal middle plane between a SiNP (radius = 70 nm) and a glass substrate (A) at the two excitation wavelengths: 532 nm (B) and 635 nm (C). The values at the different positions where the emitter is placed are summarized in Table S1. Scale bar: 50 nm.

	$ E ^2/ E^0 ^2 @ \lambda = 532 \text{ nm}$	$ E ^2/ E^0 ^2 @ \lambda = 635 \text{ nm}$
d_1	1.38	1.56
d_2	1.47	1.63
d_3	0.88	1.08

Table S1. Electric field enhancement at the various emitter distances. Extracted values from the maps in Figure S1 at the different emitter distances tested here.

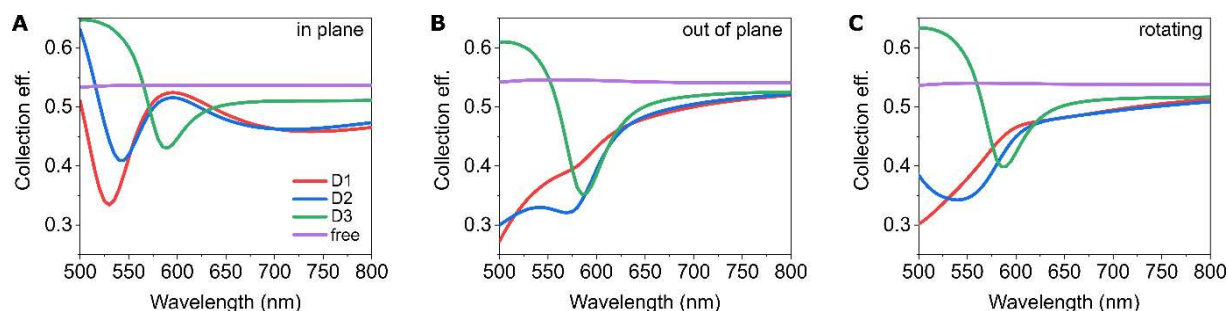


Figure S2: Collection efficiency for an emitter at different distances to a SiNP. Collection efficiency at the glass side for a dipole sitting at a glass-water interface with in plane (A), out of plane (B) or averaged (rotating) (C) orientation placed at different distances (d_1 , d_2 and d_3) from a SiNP (radius = 70 nm) as a function of its emission wavelength. The case of a dipole without any SiNP (free) is used as a reference.

	η/η^o for ATTO 542	η/η^o for ATTO 647N
d_1	0.85	0.91
d_2	0.82	0.91
d_3	0.93	0.94

Table S2. Changes in collection efficiency at the various emitter distances. Displayed values are obtained by multiplying the values shown in Figure S2 with the normalized emission spectrum of each dye and integrating over the whole spectral range.

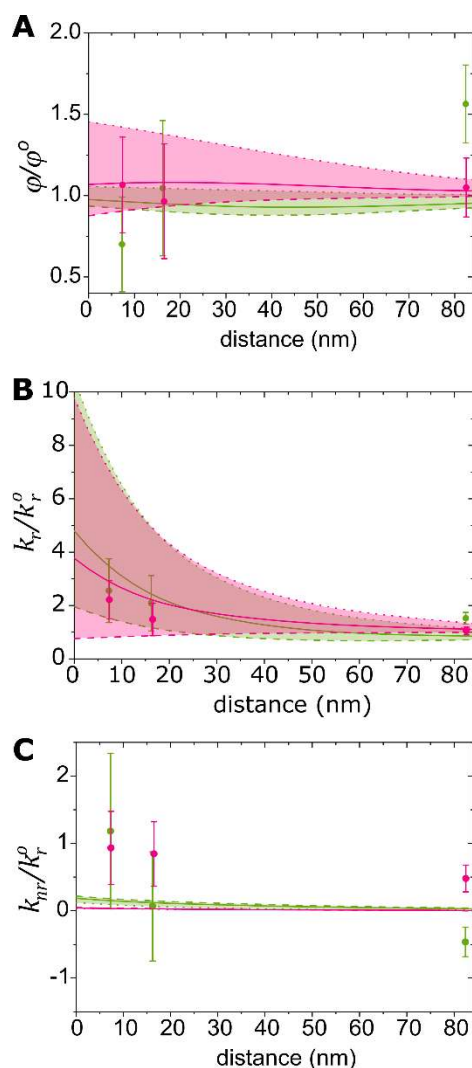


Figure S3: Comparison between experimentally extracted rates (after applying a correction factor) and simulated values. Changes in quantum yield ϕ (A), radiative decay rate k_r (B) and non-radiative decay rate k_{nr} (C) with respect to the isolated molecule as a function of the dye-SiNP distance for the two emitters: magenta for ATTO647N and green for ATTO542. Shaded areas indicate the range between the simulated values for radial (dotted) and tangential (dashed) orientations, and the solid line represents the trend for an average orientation. Experimentally obtained values (bars showing standard error around the median) fall well within those expected ranges. Intensity ratios obtained experimentally are converted into quantum yield ratios by multiplying by a correction factor that takes into account simulated changes in excitation rate and collection efficiency in the presence of the SiNP.

List of DNA staples:

Name	Sequence
Biotin1	ACAGGAAGATTGTCCCCCTTATTCACCCTCATTTGTTTC-Biotin
Biotin2	GTTGATAGATATAAGCATAAGTATAGC-Biotin
Biotin3	AGAGTACTCACGCTAACCTTTAATTGC-Biotin
Biotin4	CACTAAAACACTCACGAACTAACACTAAAGT-Biotin
Biotin5	TCACGACGTTGGGCGCTTTGGTAAAAC-Biotin
Biotin6	CAGAGATAGCGATAGTGAATAACATAA-Biotin
ATTO542_P1	AATTGAGTAATATCAGAAAATAAACAGCCATA-ATTO542
ATTO647N_P1	TACCCCGGTTAAAATTCCTTTGCCCGAACGTT-ATTO542
ATTO542_P2	ATTO542-CCCCGCTGAGAGCCAGCAGGCCTGCAACAGTGCCACCCC
ATTO647N_P2	ATTATCACCGGAAATGTTAGCAAACGTAGAA-ATTO647N
ATTO542_P3	ATTATTACTTGGGAAGTTCATTACCCAAATCA-ATTO647N
ATTO647N_P3	ATTO647N-CCCCAGAGCGGGAGCTAACTTTCCTCGTTAGAATCCCC

Table S3: List of biotinylated and dye-modified staple sequences used in this project compared to the original design.^[47] ATTO 647N and ATTO 542 staples present a single ATTO 647N or ATTO 542 fluorescent molecule for the alternative positions (P1, P2 and P3), and Biotin staples allow binding to a neutravidin-functionalized glass surface.

Name	Sequence
H.left 1(A8)	TACAAATTGCCAGTAAAGTAATTCTGTCCAGAAAAAAAA
H.left 2(A8)	TAAAGACTGTTACTTAGGCGCAGACGGTCAATAAAAAAAA
H.left 3(A8)	GCCACCCCTTCGATAGCATAATCCTGATTGTTTAAAAAAAA
H.left 4(A8)	CACCAACCAAGTACAAGTACAGACCAGGCGCAAAAAAAAA
H.left 5(A8)	CGACATTCCCAGCAAAATTATTTGCACGTAAAAAAAAAAA
H.left 6(A8)	CTTTTTTTCATTCAACAATAACGGATTTCGAAAAAAAA
H.left 7(A8)	ACGGGTAATAAATTGTTGACCAACTTTGAAAGAAAAAAAA
H.left 8(A8)	ATTAATTATGAAACAATATACAGTAACAGTACAAAAAAAA
H.left 9(A8)	AATATTGACGTCACCGTGCGTAGATTTTCAGGAAAAAAAA
H.left 10(A8)	AACAGTAGCCAACATGACATGTTTCAGCTAATGAAAAAAAA
H.left 11(A8)	AAGAACGCAAGCAAGCATAATATCCCATCCTAAAAAAAAA
H.left 12(A8)	CTTGCGGGGTATTAATAAACCAATCAATAATCAAAAAAAAA
H.left 13(A8)	AAATCAGATCATTACCATCAACAATAGATAAGAAAAAAAA
H.left 14(A8)	CATATGGTAGCAAGGTAATGGAAGGGTTAGAAAAAAAAA
H.left 15(A8)	CCATCTTTCGTTTTCAAACCACCAGAAGGAGCAAAAAAAAA
H.left 16(A8)	GAGCCACCATCAAGTTTCTGATTATCAGATGAAAAAAAA
H.left 17 (A18)	AAAAAAAAAAAAAAAAAAGGATTATACTTCTGAACCGGAAACGTCACCAA
H.left 18 (A18)	AAAAAAAAAAAAAAAAAACCTACCATATCAAAATCACCAGTAGCACCA
H.left 19 (A18)	AAAAAAAAAAAAAAAAAATTTACGAGCATGTAGCCAAGTACCGCACTCA
H.left 20 (A18)	AAAAAAAAAAAAAAAAACATAAGGGAACCGAACGTCGAAATCCGCGACC
New Core 1	GAAACGCAAAGACACCGCCAAAGATACCGAAG
New Core 2	ATTTTGCACCCAGCTATTAGCGAAAGAATTAA
New Core 3	AGCGAAAGACAGCATCAGGAAGTTTGTAGCAT
H.left 22(A18)	AAAAAAAAAAAAAAAAAAGCCGCCACTCATCAATAGCACCGTAATCAGT
H.left 23(A18)	AAAAAAAAAAAAAAAAAATGGCAATCAGAACCCGCCTCCCTCAGGAGG
H.left 28(A8)	TGAGCGCTTAAGCCCATGGCATGATTAAGACTAAAAAAAA
H.left 29(A8)	CTGAACACATAGCAATAAACGCAATAATAACGAAAAAAAA
H.left 30(A8)	AAACAGGGAAGAAAAGGCCGAACAAAGTTACAAAAAAAA
H.left 31(A8)	TCCACAGAGGTGTATCGGATAAGTGCCGTCGAAAAAAAA
H.left 32(A8)	GTCACCAGCGCCACCCTTAGGATTAGCGGGGTAAAAAAAA
H.left 33(A8)	TAGGAACCCACCCCTCTATTAAGAGGCTGAGAAAAAAAA
H.left 34(A8)	AATTAGAGAACCGATTGGCAACATATAAAAAAAAAAAA
H.left 35(A8)	TTACCATTTTACCAGCACGGAATAAGTTTATTTAAAAAAAA
H.left 36(A8)	TGAAACCACAGAACCGAGCCACCACCCTCAGAAAAAAAA
H.left 37(A8)	GCGACAGAACCGGAACACCACCAGAGCCCGCAAAAAAAAA

Table S4: List of DNA modified staples, which were extended as NP capturing strands, used for the directional antennas compared to the original origami design.^[47] Handles are staples extended with an PolyA (number of A indicated in parenthesis) to bind a single PolyT functionalized SiNP.